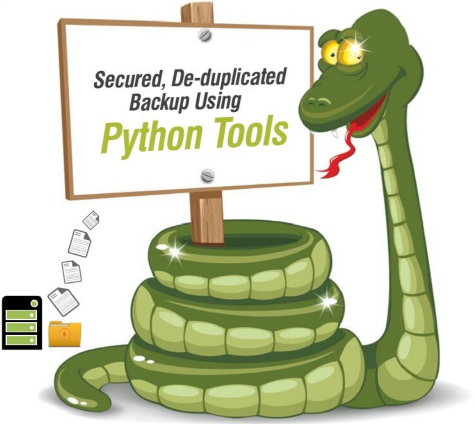

Critical data on any system can be safeguarded against accidental erasure, corruption, damage, etc, by an efficient backup system. Backup and recovery are integral to any well-managed computer system, be it for personal computing or enterprise computing. Explore the essentials of a backup system in this article and take a tutorial on the Python tools, Bakthat and Attic.

Backup and recovery of heterogeneous and unstructured data is a time consuming process. A number of software products and tools have been developed for this purpose, yet the backup of files, folders and repository maintenance is an ongoing process for higher efficiency, integrity, security and de-duplication of content.

Whenever routine office or personal work is done on our laptops or systems, only a small percentage of data on the full hard disk is modified. Let's suppose there is 500GB of hard disk space in a laptop. After regular office or personal tasks, some of the files are changed. Some files and documents can be downloaded from the Internet and stored on the local system. This means there is a change of only a few MBs per day. But when we take a backup, all the drives are copied to a portable hard disk or any other storage medium. This type of backup, which is very time consuming and not implemented on a regular and frequent basis, is known as a 'full backup'.

There are a number of mechanisms by which the modified or newly created files can be stored in your system in very little time.

First of all, let's get familiar with the basic taxonomy of the backup paradigm. Broadly, there are three types of backups in classical computing for multiple domains.

First, there is differential backup, during which only the data that was not present in the last *full backup*, is stored and saved. If a full backup is taken on July 1, the differential backup on July 2 will copy only the files changed since July 1. All other files will not be copied. Such a backup system is more flexible, efficient and very fast compared to full backups. Differential backup is also known as cumulative incremental backup.

Incremental backup, the second type, supports the copying of files that were modified since the last *backup* —whether it was a differential backup or full backup. For example, if a full backup was done on July 27, 2015, the incremental backup on July 28 will copy all the files which were changed since July 27, 2015. Similarly, if incremental backup is performed on the next day, only the changed files will be copied. This is also known as differential incremental backup.

The third variety, full backup, is performed periodically, like once a week or twice a month. This type of backup is done when major changes on the disk are performed, which can include a software install/uninstall, an upgrade or any such similar task. In a full backup, all the files are fetched, but this process is very time consuming. If any segment of the disk does not have any data, even the empty backup from that location is taken, which leads to the consumption of processor time and overhead.